

Puzzles Bodies

1. **Position**: The (x, y) coordinates of the body's center within the scene.

2. **Body Type**: Determines the physics behavior of the object:

1. **Static** (0): Fixed objects that do not move during simulation.

2. **Dynamic** (1): Objects that respond to forces and collisions.

3. **Angle**: The rotation of the body in radians.

4. **Relationship**: Define the goal condition

5. **Color:** An integer index that maps to a specific color:

1. Red (0)

2. Black (1)

3. Green (2)

4. Azure/Blue (3)

5. Purple (4)

6. Grey (5)

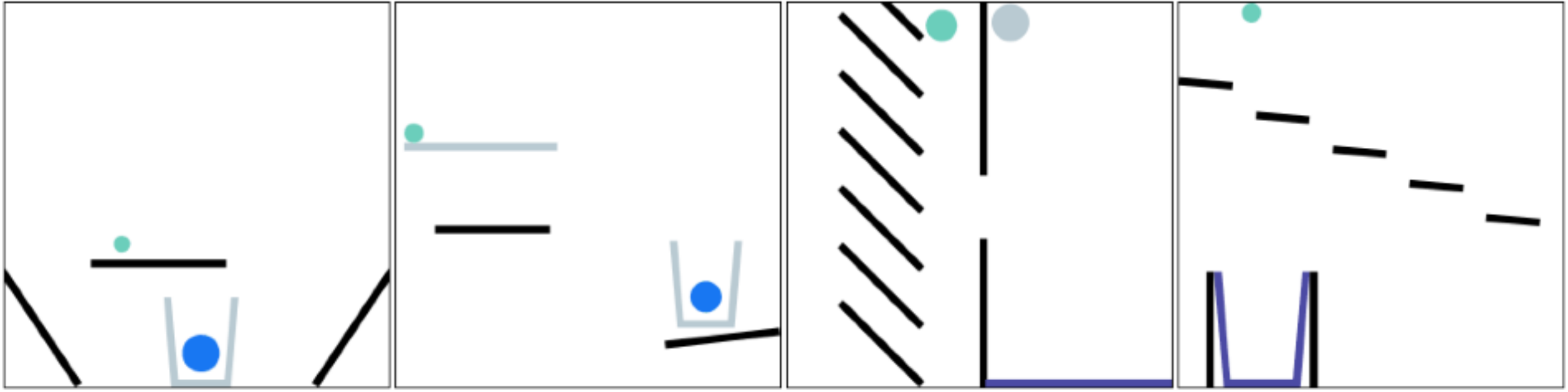
6. **Shape Type**: Defines the geometric form of the body:

1. **Polygon** (0, 2): Defined by a set of vertices.

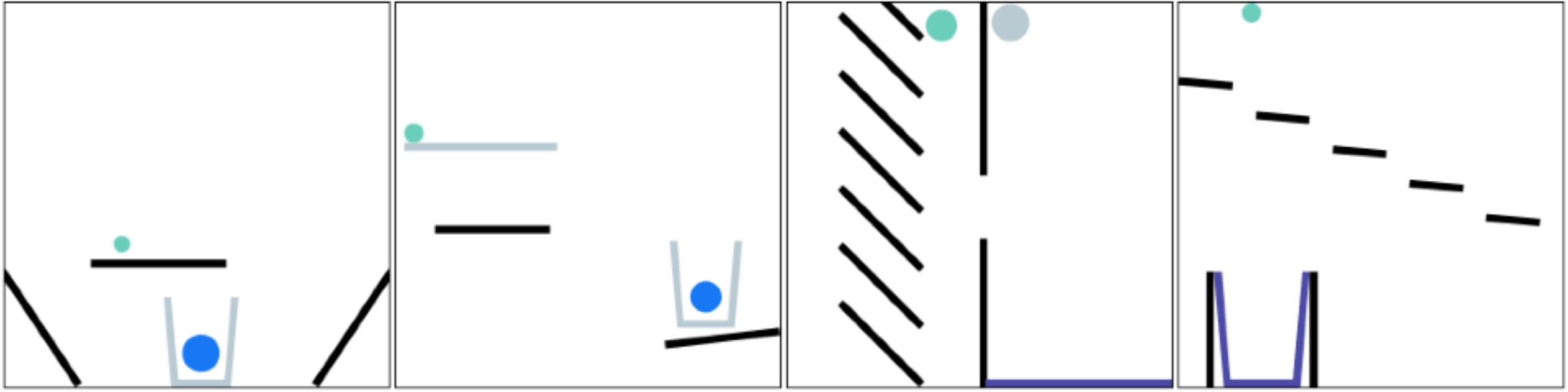
2. **Circle** (1): Defined by a center point and radius.

3. **Compound** (3, 4): Complex shapes composed of multiple polygons.

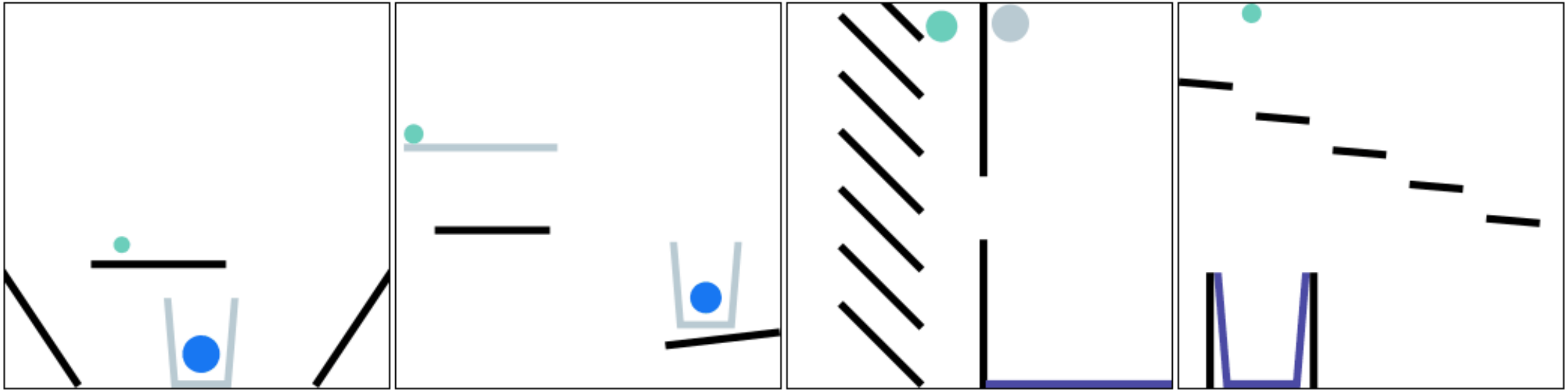
Make the green ball touch the blue/purple object by adding red objects



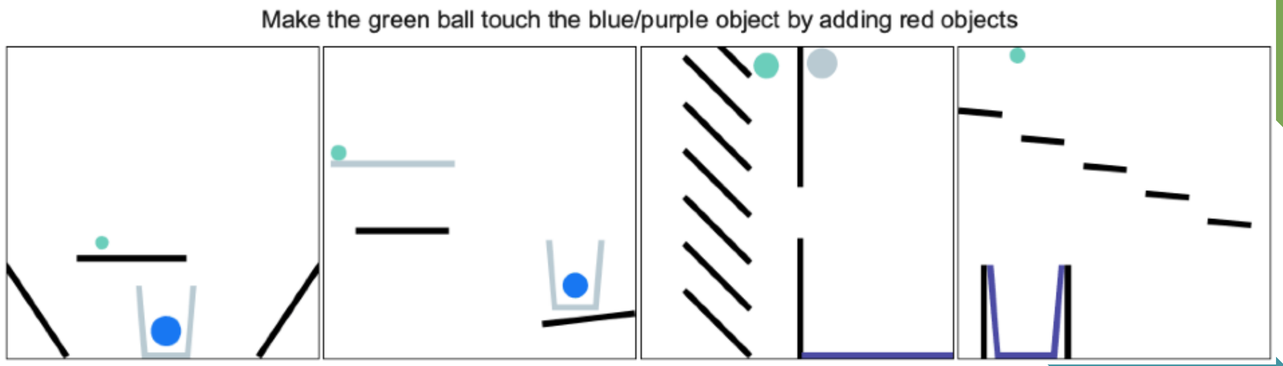
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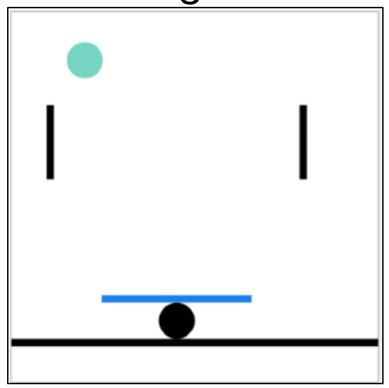
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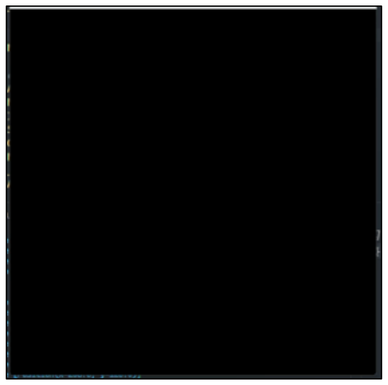


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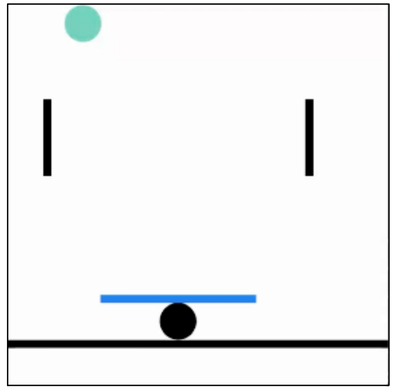
Simulation Constants

Original





Pymunk



PyBox2D

Category	Parameter	Description
Physical	Scene dimensions	Size of the simulation scene (256 × 256 pixels)
	Gravity	Acceleration due to gravity (9.81 m/s^2)
	Density	Mass per unit area (0.25 kg/m^2)
	Friction coefficient	Coefficient determining resistance to sliding (0.5)
	Elasticity (restitution)	Bounciness of collisions (0.20)
	Angular damping	Reduction factor for rotational motion (0.01)
	Linear damping	Reduction factor for linear motion (0.0)
	Min Proposal Radius	Minimal length of the proposals (2)
	Max Proposal Radius	Maximum length of the proposals (32)
Simulation	Frame rate	Number of frames per second (60 FPS)
	Time scale	Factor to adjust simulation speed (1.0)
	Scene dimensions	Rendered scene dimensions (256 × 256 pixels)
	Velocity iterations	Iterations for the velocity solver (10)
	Position iterations	Iterations for the position solver (10)
Stopping	Stop velocity threshold	Threshold below which an object is considered static (0.1)
	Required frames for early stop	Consecutive frames with static objects to trigger early stopping (400)
	Required frames for goal verification	Consecutive frames where the two target objects had to remain in contact (360)
	Max frames	Maximum frames allowed for simulation before forced termination (3000)